SEPARATEDNESS CHECKED ON AN ELEMENTARY DISTINGUISHED SQUARE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3218-3224 of sources/stacks-spaces-pushouts/pinned-original-source.tex; proof span is lines 3226-3237, the complete author proof (10 body lines).

Editorial note. Navigation stubs: 0 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 37 of queue_spaces-pushouts.json (pb=579, stmt 3218-3224, proof 3226-3237).

Editorial note. spaces-pushouts wave-1: fresh adjudication accept (2026-10-14); six-check dispatched pool (§2 row appended 02:4x).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Lemma 0.1. *Let S be a scheme. Let $X \to Y$ be a morphism of algebraic spaces over S. If $(U \subset X, f : V \to X)$ is an elementary distinguished square such that $U \to Y$ and $V \to Y$ are separated and $U \times_X V \to U \times_Y V$ is closed, then $X \to Y$ is separated.*

Proof. We have to check that $\Delta : X \to X \times_Y X$ is a closed immersion. There is an étale covering of $X \times_Y X$ given by the four parts $U \times_Y U$, $U \times_Y V$, $V \times_Y U$, and $V \times_Y V$. Observe that $(U \times_Y U) \times_{(X \times_Y X),\Delta} X = U$, $(U \times_Y V) \times_{(X \times_Y X),\Delta} X = U \times_X V$, $(V \times_Y U) \times_{(X \times_Y X),\Delta} X = V \times_X U$, and $(V \times_Y V) \times_{(X \times_Y X),\Delta} X = V$. Thus the assumptions of the lemma exactly tell us that Δ is a closed immersion. □

Date: stacks-project:spaces-pushouts.tex:local-snapshot-20261012; GNU FDL 1.2 or later.